

Introduction to Greedy Methods

Making Change – A Problem

- It is a problem of finding a change for some amount of rupees, using infinite supply of notes/coins of different valid denominations .

- **Example:**

A customer is buying an Item from a shop which costs him ₹135, for which he gives a note of a ₹500 note. So now shopkeeper needs to give back ₹365. Find a change for the amount ₹365 using valid denominations.

Making Change – Possible Solutions

Denominations	Solution 1	Solution 2	Solution 3	Solution 4
200	0	0	1	1
100	3	3	1	1
50	0	0	1	0
20	2	3	0	3
10	0	0	1	0
5	1	0	1	0
1	0	5	0	5
# Notes / Coins	6	11	5	10

Making Change – Optimal Solution

- All the solutions shown in the previous slide are called **feasible** solutions
- But if it is required to give change in minimum possible number of notes/coins then #3 is the solution.
- Here the above condition makes us to give the change using minimum number of notes/coins and hence it pushes us to find the **optimal** solution.

Optimization Problem

- Optimization problems are those problems in which aim is to **minimize** or **maximize** some cost function.
- The cost function that is being maximized or minimized is known as **objective** function.
- The process of maximization or minimization often involves a range of choices. The goal of optimization is to select the best possible choice under the given criteria.

Optimization Problem

- Various possible choices for a solution are referred as **feasible** solutions.
- Amongst the feasible solutions, the one which maximizes/minimizes the objective function is called an **optimal** solution.
- When maximization/minimization is required to be achieved under some constraint, the problem is called **Constrained Optimization**. (Optimization without any constraint is known as 'unconstrained optimization').

Definition - Optimization Problem

- “Given a function $f : A \rightarrow \mathbb{R}$ from some set A to the real numbers and a function $g : A \rightarrow \{0, 1\}$, search an element x_0 in A such that $f(x_0) \geq f(x)$ for all $x \in A$ and $g(x_0) = 1$.”
- In mathematical formulation this is referred as

$$\operatorname{argmax}_x f(x)$$

which means “those values of x for which $f(x)$ is maximum”.

Definition - Optimization Problem

- **Objective function** : The function which we aim to maximize. In above definition, $f(x)$ is the objective function.
- **Constraint** : The condition that x must satisfy. In above definition, $g(x)$ is the constraint.
- **Feasible Solution** : All solutions for which $g(x_0) = 1$.
- **Optimal Solution** : A feasible solution which maximizes $f(x)$, $\operatorname{argmax}_x f(x)$

Combinatorial Optimization

- An optimization problem with discrete variables is known as a combinatorial optimization problem.
- In a combinatorial optimization problem, we are looking for a solution from a finite (or possibly countable infinite) set of feasible solutions.
- In other words, a set of feasible solutions is discrete and aim is to find the best solution.

Combinatorial Optimization

- A brute-force approach to solve any combinatorial optimization problem is to enumerate all feasible solutions and then select the best amongst them.
- For a given input set S of size n , generating all possible subsets of feasible solution takes $O(2^n)$ time i.e. exponential time.
- Thus, for a large input it is very difficult to generate all solutions and find the optimal solution.
- In such cases **Greedy method** can be used to generate optimal solution without exploring all feasible solutions.

Greedy Method

- A greedy algorithm for optimization problem usually starts with an empty solution set and with each iteration generates a partial solution by adding one element into the set.
- It is necessary to ensure that the (partial) solution generated after each iteration is a feasible solution.
- A selection function needs to be defined to add an item during each iteration.
- Greedy method ensures that the selection made in each iteration is the best choice.

Greedy Method

Data: $a[1:n]$ contains n inputs

solution = Φ ;

for $i \leftarrow 1$ **to** n **do**

$x[i] = \text{Select}(\text{next best candidate from the set } a[1 : n])$;

if $\text{Feasible}(x[i], \text{solution})$ **then**

$\text{solution} = \text{Union}(\text{solution}, x[i])$;

return solution ;

General structure of a greedy method

Greedy Method

- Basic underlying assumption in the greedy method is,

‘If the subsolution at each iteration is optimal, then the final solution is also optimal’.

- However, greedy method does not always guarantee an optimal solution.
- Greedy method is popular because it is often the most simple and intuitive approach.

Making Change - Greedy Solution

Algorithm *MakeChange* (n)

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$D = \{1, 2, 5, 10, 20, 50, 100, 200, 500, 2000\}$

$S \leftarrow \emptyset$ */* S is a set that will hold the solution */*

$s \leftarrow 0$ */* s is the sum of the items in S */*

while $s \neq n$ do

$x \leftarrow$ the largest item in D such that $s + x \leq n$

 if there is no such item then

 return "no solution found"

$S \leftarrow S \cup \{ \text{a denomination of value } x \}$

$s \leftarrow s + x$

return S

}

Making Change - Time Complexity

- For sorting m coins $O(m * \log(m))$.
- While loop, the worst case is $O(n)$. If all we have is the coin with 1-denomination.
- Complexity for coin change problem becomes $O(m * \log(m)) + O(n)$.

References

1. A Book – “**Fundamentals of Computer Algorithms**” by Horowitz, Sahni, Rajasekaran
2. A Book – “**Fundamentals of Algorithms**” by Brassard & Bratley